Prompt:
You are a Data Analyst, tasked with creating an interactive dashboard to analyze the data in NR_dataset.xlsx. Use the fields label, purchaseamount, customerregion, productcategory, retailchannel, customerid, and transactiondate to create this dashboard. The dashboard should help identify growth opportunities, detect early warning signs of customer decline, and support strategic decisions. The dashboard should support the identification and communication of actionable insights to guide decision-making, including:

1. Identification of potential and emerging growth opportunities in sales within the existing customer base.
2. Detection of early warning signs of customer decline or reduced engagement.
3. Recommendation of data-driven actions to support commercial and marketing initiatives, helping define strategic priorities for revenue optimization.

Create this dashboard in python, name it app.py, and create it to be deployed through GitHub and Streamlit. Prepare all required files for GitHub deployment, including the dataset, requirements.txt file, and the Python application file.

Explanation:
For this prompt, I made sure to use the role based prompting approach, by starting off with "You are a Data Analyst". For the fields, I wanted to limit the scope of the dashboard to only fields that matter for Sophia's analysis. Therefore, I dropped Gender, Satisfaction, and Age Group from my analysis. I considered dropping Transaction Date as well, but I thought that time based analysis would still be useful to "detect early warning signs of customer decline". Initially, I had not included the fields in the prompt, which led the output to look extremely messy and hard to parse. The filters were not something that I had asked for, but they ended up looking pretty good, so I decided to leave them in.

I had to do a few things after the prompt to clean up the data and dashboard. First, I deleted Row 99 as it didn't have a classification between Customer Segments. I also had to lightly edit the Strategic Recommendations, as the initial output had far too many recommendations, and I wanted to limit it to around 5-6. I settled on two growth, two retention and two strategy themed recommendations. One more small issue was that the Strategic Recommendations had a small visual error in the formatting, where two dollar signs ($) in the same line of text would be considered a LaTeX math delimiter. Changing any instance of "$" that had this problem to "\$" solved the issue. This dashboard will only pull data from NR_dataset.xlsx, but a good improvement would be the ability to import any data set with the same format. A much larger dataset would be very valuable to see if the trends we see in this data continues.